

Part I. Calculus I competency.

1. (5 pts) Find the equation of the tangent line to the graph of the function $f(x) = \frac{5}{(x-1)^2}$ at $x = 3$.
2. (5 pts) Find the derivative of $a(x) = x^{3/5} - \frac{2}{x^4}$
3. (5 pts) Find the derivative of $b(x) = \frac{4x+5}{3-2x}$
4. (5 pts) Find the derivative of $c(x) = e^{-3x} \cos 4x$
5. (5 pts) Find the derivative of $e(x) = x - x \ln x$
6. (5 pts) Find the derivative of $f(x) = (x^2 - x + 1)^4$
7. (5 pts) Find the indefinite integral $\int (\sqrt{x} - 3x^3) dx$
8. (5 pts) Find the definite integral of $\int_0^{\pi} (1 - \sin x) dx$
9. (5 pts) Find the indefinite integral $\int (3 - 2x)^4 dx$
10. (5 pts) Find the indefinite integral $\int \frac{(\ln x)^2}{x} dx$

Part II. Calculus II Competency.

11. (5 pts) Find the area bounded between the graphs of $y = x^3$ and $y = x$.
12. (5 pts) Find the indefinite integral $\int x \cos x dx$
13. (5 pts) Find the partial fraction decomposition for the function $f(x) = \frac{3x-1}{x^2-4}$
14. (5 pts) Find the indefinite integral $\int \frac{2x-1}{(x-x^2)^2} dx$
15. (5 pts) Test the series for convergence: $\sum_{k=1}^{\infty} \frac{1}{k^3 + k}$
16. (5 pts) Test the series for convergence: $\sum_{k=1}^{\infty} k^2 \left(\frac{5}{4}\right)^k$
17. (5 pts) Find the first four non-zero terms of the MacLaurin series of $f(x) = \sin(x^2)$

Part III. Calculus III competency.

18. (5 pts) Find f_x, f_y for $f(x, y) = ye^{-2xy}$
19. (5 pts) Find f_{xx}, f_{yx} for $f(x, y) = ye^{-2xy}$
20. (5 pts) Find f_x, f_y, f_z for $f(x, y, z) = \frac{xz + y^2}{z^3}$